

SCX Theory: Submission-Ready Checklist

for arXiv

Date: 2026-06-28 | **Status:** Final pre-submission audit **Purpose:** Complete inventory of theorems, assumptions, notation, and author notes for arXiv submission.

1. Complete Theorem Inventory

1.1 Core Theorems (Proven)

#	Name	Status	Key Equation	Page Estimate
Thm 1	Multi-Expert Consistency Guarantee for Label Noise Detection	PROVEN	$F1 \geq 1 - \frac{1}{\eta} \sum_s \rho_s e^{-2M\Delta_s^2}$	8-10 pages

#	Name	Status	Key Equation	Page Estimate
Thm 2	Weak Feature Failure Lower Bound	PROVEN	$F1_{\text{SCX}} \leq F1_{\text{base}} + C_F \sqrt{1 - e^{-\delta}}$	6-8 pages
Thm 3	Noise-Difficulty Uniden-tifiabil-ity	PROVEN	$\mathcal{P}_{\text{noise}} = \mathcal{P}_{\text{hard}}$ constructively	8-10 pages
Prop 3	State-Conditioned Expert Weight-ing	PROVEN	$\Delta R \geq \frac{1}{\alpha} I(s; w)$	3-4 pages
Prop 4	Compression Fidelity (SCX-Compress)	PROVEN	$\varepsilon(s) \leq B(1 - D_{\text{eff}})(1 - n'_s/N_s) + O(B\sqrt{d/n'_s})$	5-7 pages
Thm 12.5	Lyapunov Descent Under Reference-Set Replay	PROVEN	$\mathbb{E}[\Delta\Phi_t] \leq -\alpha_t \ \nabla L_0\ ^2 - 2\beta_t \rho \ \Delta_t\ \ S_t - \hat{C}\ + O(\alpha_t^2 W + \beta_t^2)$	4-5 pages

#	Name	Status	Key Equation	Page Estimate
Thm 12.2	Impossibility of Lya-punov Descent Without Replay	PROVEN	Requires $D_{\text{joint}}^* + 2B(1 - \varepsilon) = 0$	2 pages
Spring 1	Convergence of SCX (SE-Self-Evolution)	PROVEN (with Thm 12.5)	$(S_t, \theta_t) \rightarrow (S^*, \theta^*)$ a.s.	6-8 pages
Spring 2	Completeness Bound (SE-Finite-Time Termination)	PROVEN	$T^* \leq \Phi_0 / \varepsilon_{\text{mach}}$	4-5 pages

1.2 Supporting Results (Proven)

#	Name	Status	Role
Lemma 1 (Thm 1)	Mean Separation	PROVEN	Establishes $\mathbb{E}[C \parallel \text{noise}] > \mathbb{E}[C \parallel \text{clean}]$
Lemma 2 (Thm 1)	FPR Upper Bound (Hoeffding)	PROVEN	$\text{FPR}_s \leq e^{-2M(\theta - \mu_s)^2}$

#	Name	Status	Role
Lemma 3 (Thm 1)	TPR Lower Bound (with A6)	PROVEN	$\text{TPR}_s \geq 1 - e^{-2M(1-C_{\text{bal}}\mu_s/(K-1)-\theta)^2}$
Lemma 1 (Thm 2)	State Estimation Error (Fano)	PROVEN	$P(\hat{S} \neq S) \geq (H(S) - \delta - \log 2)/\log K_S$
Lemma 2 (Thm 2)	SCX Reliability Degradation	PROVEN	$\mathbb{E}[C(\hat{S}) - C(S)] \leq 2P(\hat{S} \neq S) + O(1/\sqrt{n_{\min}})$
Lemma SE-1.1	Lyapunov Non-Increase (on training distribution)	PROVEN	Supermartingale argument
Lemma SE-1.2	Finite Memory Bank Configurations	PROVEN	Covering number + monotonicity
Lemma SE-1.3	Vanishing Parameter Displacement	PROVEN	$\ \theta_{t+1} - \theta_t\ \leq \alpha_t G \rightarrow 0$
Lemma SE-1.4	Limit Points Are Fixed Points	PROVEN	From SE-1.1 through SE-1.3
Lemma 12.1	Domain Adaptation Bound for D_{static}	PROVEN	Ben-David et al. (2010) framework
Lemma 12.2	Alignment Bias Decomposition	PROVEN	Cauchy-Schwarz + Lipschitz

1.3 Conjectured / Open

#	Name	Status	Reason
Regime 3	Perpetual Discovery Rate	CONJECTURED	Requires tail characterization of data distribution
Phase Dia-gram	Regime Boundaries	CONJECTURED	Phase transitions not rigorously characterized
Full Func-tional Conver-gence	Infinite $\mathcal{X}$ limit of SE-1.2	OPEN	Requires Arzelà-Ascoli compactness; ε -precision version proven
G5 ($K > 2$ Thm 3)	Construction uses “completely random experts”	PROVEN but extreme	$K = 2$ construction is the practically relevant case

2. Complete Assumption Inventory

2.1 Assumptions A1-A6 (Theorem 1)

ID	Assumption	Formal Statement	Justification	Verifiability
A1	Disjoint training sets	$D_m \cap D_{m'} = \emptyset,$ $D_m \perp D_{m'}$	Ensures expert errors are independent; breaks Thm 3's unidentifiability (Corollary 2)	Testable: verify training data provenance
A2	Clean-data conditional independence	$\{e_m(x, y)\}_m$ conditionally independent given x	Follows from A1; required for Hoeffding/Chernoff concentration	Implied by A1
A3	Bounded loss	$\ell(\cdot, \cdot) \in [0, B],$ $B < \infty$	Technical requirement for Hoeffding inequality	Always satisfied for bounded losses (0-1, truncated MSE)
A4	Uniform independent noise	Noise independent of x , uniform over $\mathcal{Y} \setminus \{y^*\}$	Breaks Thm 3's unidentifiability (Corollary 3); standard label noise model	Testable: χ^2 test for noise rate constancy across states
A5	State homogeneity	$\sup_{x \in s} \mathbb{E}[C \text{clean}, x]$ μ_s	Breaks Thm 3's unidentifiability (Corollary 4); enables state-level analysis	Testable: KS test for within-state error uniformity

ID	Assumption	Formal Statement	Justification	Verifiability
A6	Balanced error distribution	$\max_{c \neq y^*} \mu_c(x) \leq C_{\text{bal}} \cdot \mu_s / (K - 1)$	Breaks Thm 3's unidentifiability (Corollary 5); prevents error concentration	Testable: χ^2 test for error class balance; C_{bal} estimable

Recommended C_{bal} defaults: $C_{\text{bal}} = 2$ (conservative) or estimate from validation data.

2.2 Conditions C1-C9 (Self-Evolution, Spring-1)

ID	Condition	Formal Statement	Justification	Dependencies
C1'	Finite covering dimension	$\dim(\Phi) = d_\phi < \infty$	Replaces unrealistic “finite $\mathcal{X}$ ”; enables covering-number argument for memory bank stabilization	Always true for $\mathbb{R}^{d_\phi}$ features
C2	Lipschitz student	$\ f_{\theta_1}(x) - f_{\theta_2}(x)\ \leq L_f \ \theta_1 - \theta_2\ $	Required for gradient control in SGD analysis	Standard for neural networks with Lipschitz activations

ID	Condition	Formal Statement	Justification	Dependencies
C3	Lipschitz gatekeeper	$\ \text{SCXUpdate}(S_1) - \text{SCXUpdate}(S_2)\ _\infty \leq L_S \ S_1 - S_2\ _\infty$	Required for gatekeeper update control	Holds for averaging-based SCX updates
C4	Robbins-Monro rates (student)	$\sum \alpha_t = \infty, \sum \alpha_t^2 < \infty$	Standard SGD convergence condition	Canonical: $\alpha_t = t^{-a}$, $a \in (1/2, 1]$
C5	Conditional i.i.d. sampling	$(x_t, y_t) \ S_t \sim P_{S_t}$	Models acceptance-biased data generation	Follows from gatekeeper definition
C6'	Two-timescale separation	$\beta_t = o(\alpha_t)$	Student converges between gatekeeper updates; controls distribution shift	Joint scheduling: $\alpha_t = t^{-0.6}$, $\beta_t = t^{-0.8}$
C7	Bounded gatekeeper update	$\ \text{SCXUpdate} - S\ _\infty \leq B_S$	Controls per-step gatekeeper change	Enforced by projection $\Pi_{[0,1]}$
C8	Annealing threshold	$\gamma_t \rightarrow 0.5$	Prevents gatekeeper overconfidence early in training	Practical: $\gamma_t = \gamma_0 + (0.5 - \gamma_0)(1 - e^{-t/\tau})$

ID	Condition	Formal Statement	Justification	Dependencies
C9	Random exploration	ε -fraction random acceptance	Ensures all regions remain represented; bounds $\text{TV}(P_{S_t}, P_0) \leq 1 - \varepsilon$	Implementable as ε -greedy gatekeeper

2.3 Additional Assumptions (Theorem 12.5 — Reference-Set Replay)

ID	Assumption	Formal Statement	Justification
R1	Bounded importance weights	$\ w_t\ _\infty \leq W < \infty$ for $w_t(x) = P_0(x)/P_{S_t}(x)$	Required for finite variance of importance-weighted gradients; implied by C9 ($W \leq 1/\varepsilon$)
R2	Reference set availability	Fixed M_0 available for SCXUpdate computation	Practical: small holdout set curated before self-evolution

2.4 Physical Constraints (Spring-2)

ID	Constraint	Bound	Justification
P1	Finite data	$ \mathcal{D}_{\text{total}} \leq N_{\text{max}} < \infty$	Physical storage limit; always true in practice
P2	Finite precision	$\varepsilon_{\text{mach}} > 0$ (typically 10^{-16} for float64)	Hardware constraint
P3	Finite parameterization	Gatekeeper and student parameterized by $\leq d$ parameters	Always true for implemented systems

3. Notation Conflicts

3.1 Critical Conflicts (Proof-Affecting)

None found. All conflicts are presentational, not logical.

3.2 Moderate Conflicts (Reader-Confusing)